Additional Notes on the SIMION HS1 Collision Model

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1 Overview

This document provides additional notes and mathematic derivations for some of the equations and algorithms used in the SIMION HS1 collision model.

2 The Model

The HS1 collision model is based on the following physical model. An ion travels through a neutral background gas, colliding with those gas particles and scattering (elastically) according to the hard-sphere model[1] of kinetic theory. In the kinetic theory of gas, the gas particles travel in randomized velocities according to the Maxwell-Boltzmann distribution[2]. That is, the gas particle velocities in each dimension (x, y, and z) are Gaussianly (normally) distributed with standard deviation $2kT/m_{gas}$, where k is the Boltzmann constant, T is temperature in Kelvin and m_{gass} is the mass of each gas particle.

3 Collision Frequency

In a sufficiently short length in time, the ion sweeps out a cylindrical volume, meaning that it collides with all gas particles with centers inside that volume. The cross-sectional area of that cylinder (the collision cross section σ) is a semiempirical constant roughly equal to the area with diameter being the sum of the diameters of the colliding ion and gas particles.

The frequency Z of collisions between the ion and only those particles of tge background gas with speed c relative to the ion is approximately $Z = c\sigma n$ where n is the density of those gas particles (particles per volume). Note that σn is the expected number of particle collisions per distance traveled, while c is the distance traveled per time, so Z is the number of particle collisions per time.

The true frequency of collisions is the sum (integral) of collision frequencies for each possible speed c:

$$Z = \sigma \int_0^\infty c\, dn(c) = \sigma \int_0^\infty cn'(c)\, dc = \sigma n \left[\int_0^\infty cf(c)dc \right] = \sigma nc \qquad (1)$$

This shows that Z is related to the average of all c's weighted by their population densities f (Maxwell distribution), From now on, c will refer to this mean relative speed between ion and background gas.

The expected distance between ion collisions λ is called mean-free path[3]. It is simply $\lambda = c_{ion}\frac{1}{Z}$ where c_{ion} is the ion speed. Substituting for Z, we get

$$\lambda = (c_{ion}/c)/(\sigma n) \qquad (2)$$

as used in the HS1 model code along with the ideal gas law $PV = nRT$.

4 Derivation of Mean Relative Speed

The mean relative speed c between ion and background gas (used in the λ calculation above) can itself be calculated as follows:

$$c = \iiint_V |\mathbf{v}_{ion} - \mathbf{v}_{gas}| f(\mathbf{v}_{gas})\, d\mathbf{v}_{gas} \qquad (3)$$

$|\mathbf{v}_{ion} - \mathbf{v}_{gas}|$ is the relative speed between the ion and a gas particle of velocity $\mathbf{v}_{gas}$, and $f(\mathbf{v}_{gas})$ represents the probability density of gas particles with that velocity. Here, f is the three-dimensional Maxwell distribution given by

$$f(\mathbf{v}_{gas}) = (A/\pi)^{3/2} \exp(-A\mathbf{v}_{gas}^2) \text{ where } A = (m_{gas}/2kT) \qquad (4)$$

Substituting,

$$c = \iiint_V |\mathbf{v}_{ion} - \mathbf{v}_{gas}|(A/\pi)^{3/2}\exp(-A\mathbf{v}_{gas}^2)\,d\mathbf{v}_{gas} \tag{5}$$

Let $\mathbf{u} = \mathbf{v}_{gas} - \mathbf{v}_{ion}$, $d\mathbf{u} = d\mathbf{v}_{gas}$.

$$\begin{aligned} c &= (A/\pi)^{3/2}\iiint_V |\mathbf{u}|\exp(-A\mathbf{v}_{ion}^2 - 2A\mathbf{v}_{ion}\mathbf{u} - A\mathbf{u}^2)\,d\mathbf{u} \\ &= (A/\pi)^{3/2}\exp(-A\mathbf{v}_{ion}^2)\iiint_V |\mathbf{u}|\exp(-A\mathbf{u}^2 - 2A\mathbf{v}_{ion}\mathbf{u})\,d\mathbf{u} \end{aligned}$$

Convert to spherical coordinates and let $r = |\mathbf{u}|, a = |\mathbf{v}_{ion}|, \mathbf{v}_{ion}\cdot\mathbf{u} = |\mathbf{v}_{ion}||\mathbf{u}|\cos\theta$.

$$\begin{aligned} c &= (A/\pi)^{3/2}\exp(-Aa^2)\int_0^\infty\int_0^{2\pi}\int_0^\pi r\exp(-Ar^2 - 2Aar\,\cos\theta)r^2\sin\theta\,d\theta\,d\phi\,dr \\ &= (A/\pi)^{3/2}\exp(-Aa^2)\int_0^\infty r^2\int_0^{2\pi}d\phi\int_0^\pi r\exp(-Ar^2 - 2Aar\,\cos\theta)\sin\theta\,d\theta\,dr \\ &= (A/\pi)^{3/2}(2\pi)\exp(-Aa^2)\int_0^\infty r^2\int_0^\pi r\exp(-Ar^2 - 2Aar\,\cos\theta)\sin\theta\,d\theta\,dr \end{aligned}$$

Let $s = -Ar^2 - 2Aar\,\cos\theta, ds = 2Aar\,\sin\theta\,d\theta$.

$$\begin{aligned} c &= A^{1/2}a^{-1}\pi^{-1/2}\exp(-Aa^2)\int_0^\infty r^2\int_{-Ar^2-2Aar}^{-Ar^2+2Aar}\exp(s)\,ds\,dr \\ &= A^{1/2}a^{-1}\pi^{-1/2}\exp(-Aa^2)\int_0^\infty r^2(\exp(-Ar^2+2Aar) - \exp(-Ar^2-2Aar))\,dr \end{aligned}$$

To solve this integral, we use

$$\int_0^\infty r^2\exp(-Ar^2+Br)\,dr = \frac{B^2+2A}{8A^3}\sqrt{\pi A}\exp(B^2/4A)(1+\mathrm{erf}(B/2\sqrt{A}))+\frac{B}{4A^2}$$

$$\int_0^\infty r^2(\exp(-Ar^2+Br)-\exp(-Ar^2-Br))\,dr = \frac{B}{2A^2}-\frac{B^2+2A}{8A^3}\sqrt{\pi A}exp(B^2/4A)\,\mathrm{erf}(B/2\sqrt{A})$$

$$\int_0^\infty r^2(\exp(-Ar^2+2Aar)-\exp(-Ar^2-2Aar))\,dr = \frac{2Aa^2+1}{2A^2}\sqrt{\pi A}\exp(Aa^2)\,\mathrm{erf}(\sqrt{A}a)+a/A$$

Substituting,

$$
\begin{aligned}
c &= A^{1/2}\pi^{-1/2}a^{-1}\exp(-Aa^2)\cdot(\frac{2Aa^2+1}{2A^2}\pi^{1/2}\exp(Aa^2)A^{1/2}\operatorname{erf}(A^{1/2}a)+(a/A)) \\
&= (2A^{-1/2}\pi^{-1/2})\cdot\left[\frac{\pi^{1/2}}{2}(A^{1/2}a+\frac{1}{2}A^{-1/2}a^{-1})\operatorname{erf}(A^{1/2}a)+\frac{1}{2}\exp(-Aa^2)\right]
\end{aligned}
$$

Let

$$\bar{c}_{gas} = \sqrt{\frac{8kT}{\pi m}} = \frac{2}{\sqrt{A\pi}} \qquad \textit{mean gas speed} \tag{6}$$

$$c^*_{gas} = \sqrt{\frac{2kT}{m}} = \frac{1}{\sqrt{A}} \qquad \textit{median gas speed} \tag{7}$$

$$s = c_{ion}/c^*_{gas} = a\sqrt{A} \tag{8}$$

Substituting simplifies the result to that used in the HS1 model code:

$$c = \bar{c}_{gas}\left(\left(s+(2s)^{-1}\right)\frac{\sqrt{\pi}}{2}\operatorname{erf}(s)+\frac{1}{2}\exp(-s^2)\right) \tag{9}$$

This result is in agreement with [4, Ding2003]. It is also found to be in exact agreement with the result of the fairly different derivation given in [5, Tolmachev2003].

Note the following results: As $c_{ion} \to 0, c \to \bar{c}_{gas}$. Also, as $c_{ion} \to \infty, c \sim \frac{\sqrt{\pi}}{2}\bar{c}_{gas}s = c_{ion}$. Further, if $c_{ion} = \bar{c}_{gas}$, then $c \approx 1.379\bar{c}_{gas}$, which is approximately the average relative speed between the gas particles themselves ($\sqrt{2}\bar{c}_{gas}$). The above results provide a rough justification for the approximation $c \approx \sqrt{c^2_{ion} + \bar{c}^2_{gas}}$.

5 Distribution of Gas Velocities upon Collision

Upon collisions, the probability distribution for the velocity of the colliding gas particle is not necessarily Maxwell. For example, ion and gas particles traveling at the same velocity will never collide. The true distribution is given by

$$p(\mathbf{v}_{gas}) \propto |\mathbf{v}_{gas} - \mathbf{v}_{ion}| f(\mathbf{v}_{gas}) \tag{10}$$

That is, the probability of a gas particle with a given velocity to collide with the ion is proportional to the relative velocity between the two particles as well as to the density of gas particles with that velocity.

If we align our velocity frame of reference so that the ion is traveling entirely in the x direction, then $\mathbf{v}_{ion} = (v_{ion_x}, 0, 0)$. The probability density for a certain v_{gas_x} is then

$$\begin{aligned} p(v_{gas_x}) &\propto \iint |\mathbf{v}_{gas} - \mathbf{v}_{ion}| f(\mathbf{v}_{gas}) dS_{yz} \\ &\propto \iint ((v_{gas_x} - v_{ion_x})^2 + v_{gas_y}^2 + v_{gas_z}^2)^{\frac{1}{2}} \exp(-A(v_{gas_x}^2 + v_{gas_y}^2 + v_{gas_z}^2)) dS_{yz} \end{aligned}$$

Setting $c_x = (v_{gas_x} - v_{ion_x})^2$ and converting to polar coordinates with $r^2 = v_{gas_y}^2 + v_{gas_z}^2$ and $dS_{yz} = r dr d\theta$, we have

$$p(v_{gas_x}) \propto \exp(-Av_{gas_x}^2) \int_0^{2\pi} d\theta \int_0^{\infty} (c_x + r^2)^{\frac{1}{2}} \exp(-Ar^2) r d\phi \tag{11}$$

$$= \exp(-Av_{gas_x}^2) 2\pi \left[\frac{1}{4} A^{-\frac{3}{2}} (2\sqrt{Ac_x} + \sqrt{\pi} \exp(Ac_x)(1 - \operatorname{erf} \sqrt{Ac_x})) \right] \tag{12}$$

$$\propto \exp(-Av_{gas_x}^2)(2\sqrt{Ac_x} + \sqrt{\pi} \exp(Ac_x)(1 - \operatorname{erf} \sqrt{Ac_x})) \tag{13}$$

The left-hand side of the expression is a Gaussian function, and the right-hand side is a function of v_{gas_x} that causes the expression as a whole to deviate from Gaussian. As $c_x \to \infty$ (i.e. $v_{ion_x} \to \infty$), the distribution approaches Gaussian with the usual standard deviation. The right-hand side of the expression, due do the definition of c_x is symmetric around $v_{gas_x} = v_{ion_x}$, and when $v_{ion_x} = 0$, the full expression is symmetric around $v_{gas_x} = 0$. Figure 1 shows a graph of this.

Similar analysis might be done for the transverse direction (y and z).

The HS1 model does account for the $|\mathbf{v}_{gas} - \mathbf{v}_{ion}|$ factor in the probability by appplying a rejection method to Maxwell distributed velocities. The simple rejection method avoids the complexity of the derivation above.

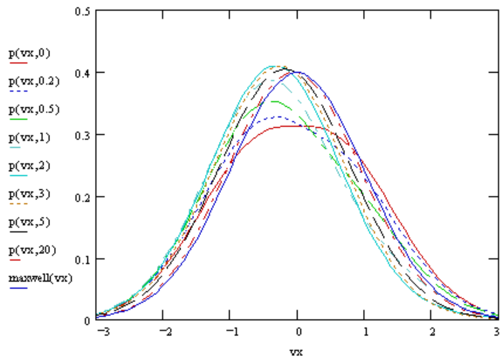

Figure 1: $p(v_{gas_x})$ for various values of v_{ion_x}, assuming $A = \frac{1}{2}$ (for one standard deviation)

6 Analysis

This section provides some validation of the HS1 model code.

Figure 2 shows the distribution of ion speeds measured at equilibrium in the HS1 model. As shown, the ions themselves become Maxwell distributed in speeds. Figure 3 is a screenshot of the ion trajectories in SIMION (it appears to roughly follow a random walk).

Figure 4 shows ion speeds v.s. collision number. The web page[6] shows a similar plot when using some other models.

References

[1] Wikipedia:Kinetic_theory http://en.wikipedia.org/wiki/Kinetic_theory

[2] Wikipedia:Maxwell-Boltzmann_distribution http://en.wikipedia.org/wiki/Maxwell-Boltzmann_distribution

[3] Wikipedia:Mean_free_path http://en.wikipedia.org/wiki/Mean_free_path

[4] Ding2002 http://dx.doi.org/10.1016/S1387-3806(02)00921-1.

[5] Tolmachev2003 http://dx.doi.org/10.1016/S1387-3806(02)00960-0

[6] http://www.simion.com/info/Collision_Model_HS1

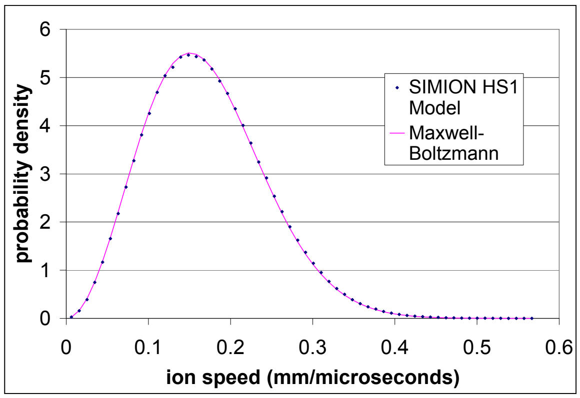

Figure 2: Distribution of SIMION HS1 ion speeds at equilibrium. Conditions: m_{ion} = 200, m_{gas} = 4 amu, σ = 2.27E-18 m^2, T=273K, P = 5 Pa. Using zero field model.

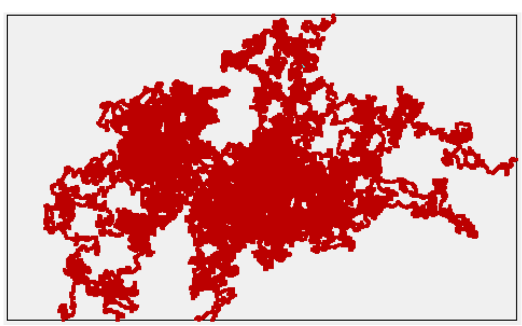

Figure 3: SIMION view of ion trajectories in previous figure.

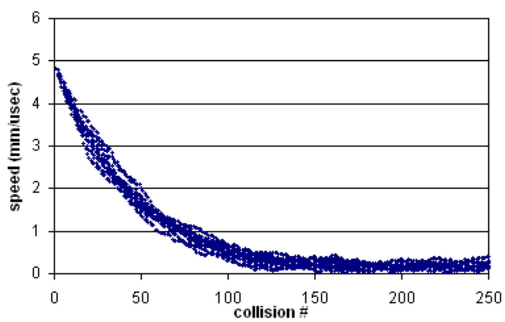

Figure 4: Damping using HS1. Conditions: ions of mass 200 amu, 15 angstrom collision diameter, and initial velocity to the right at 24 eV colliding against a He background gas with 2 angstrom collision diameter, 4 mTorr pressure, and 273 K temperature. σ = 2.27E-18 m^2.